

MediVan AI Navigation

Simulation Run Report

Duration: 71.6 seconds | Frames: 328

Distance: 1.18 meters

Camera: Real Webcam (Laptop)

Mode: SLAM Mapping (Phase 1)

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1. Session Overview

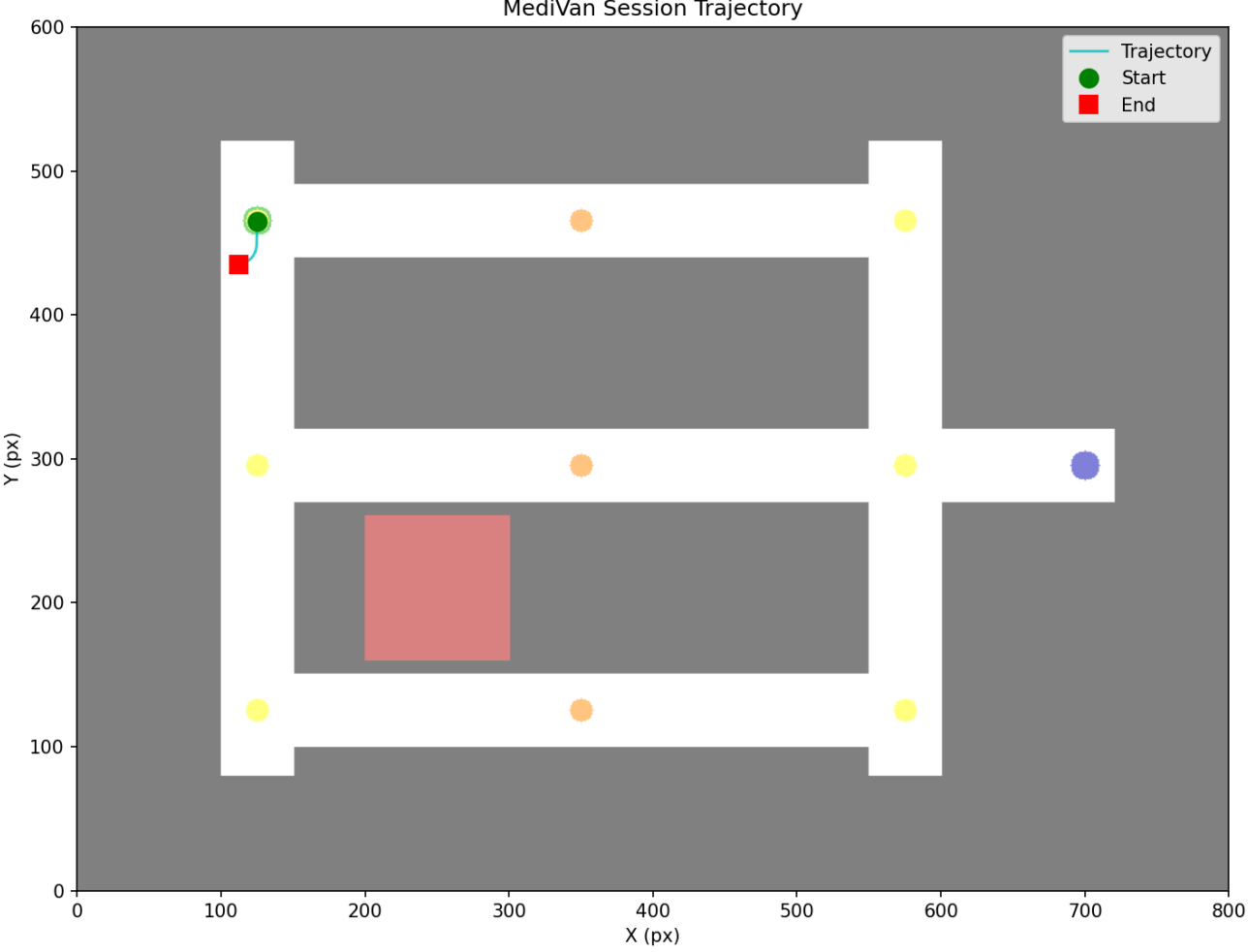
Metric	Value
Total frames	328
Duration	71.6 seconds
Effective FPS	4.6
Camera source	Real Webcam (Laptop)
AI detector mode	Heuristic classifier
Simulation mode	SLAM Mapping (Phase 1)
Operating mode	100% Autonomous

2. Movement & Navigation

Metric	Value
Position X range	111 - 125 px
Position Y range	435 - 465 px
Total distance	47 px (1.18 m)
Average speed	0.143 m/s
Max speed	0.318 m/s
Junction snaps	65
Path replans	0

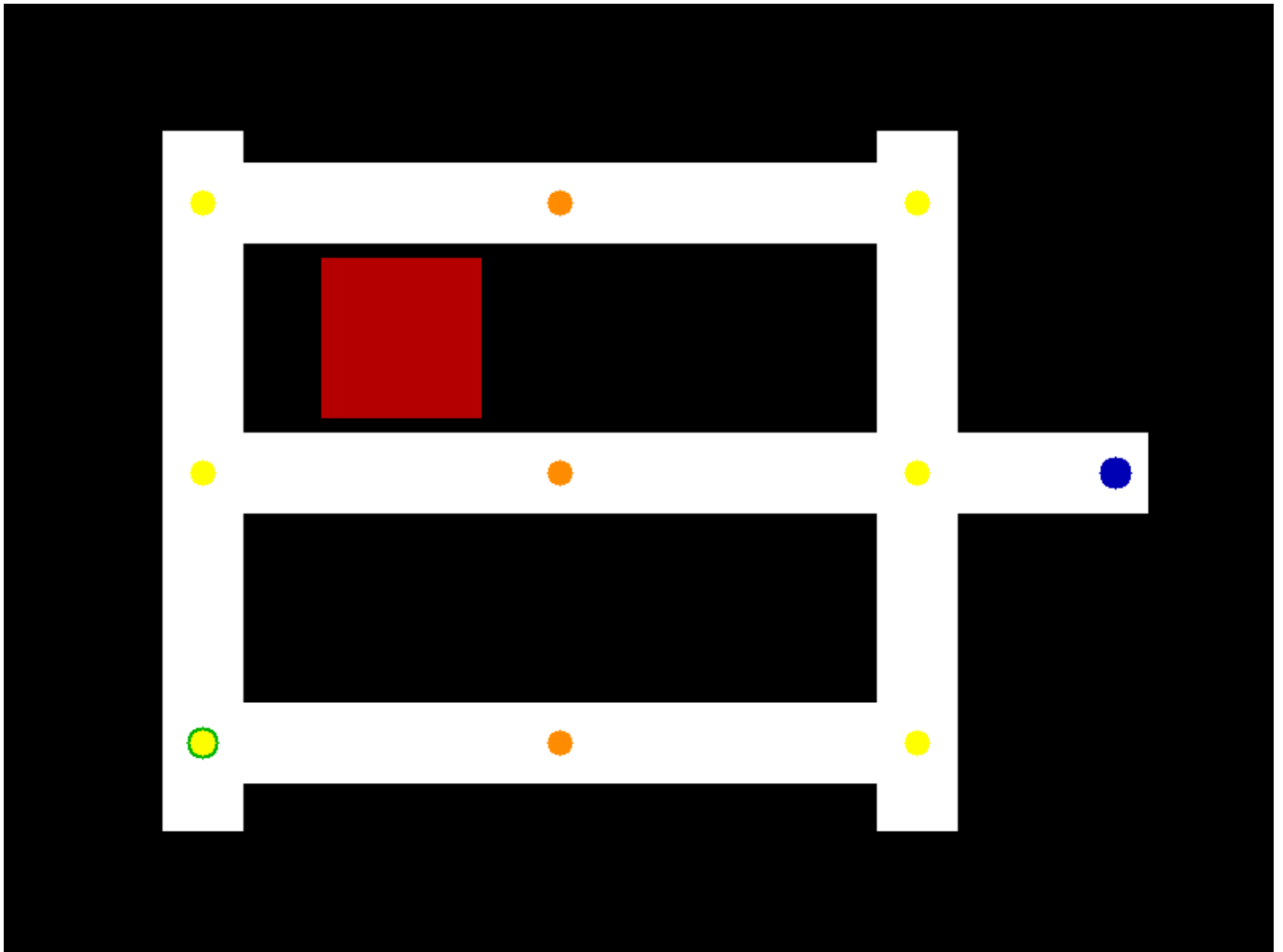
3. Trajectory Plot

The robot's path during SLAM exploration overlaid on the hospital map:



4. Hospital Map (Ground Truth)

This is the ground-truth world used for simulation physics. The robot does NOT see this map -- it builds its own via SLAM.



5. AI Obstacle Detection

Metric	Value
Frames with obstacles	328/328 (100.0%)
Max simultaneous	2

Obstacle Actions Breakdown

Action	Frames	Percentage
stop	328	100.0%

6. Battery Status

Metric	Value
Start level	100.0%
End level	99.0%
Total drain	0.98%
Drain rate	0.82%/min

7. Safety Systems

System	Events	Status
Front bumps	8	OK
Rear bumps	0	OK
Left bumps	0	OK
Right bumps	0	OK
Tilt faults	0	PASS
Wall contacts	0	PASS

8. Dock State Distribution

State	Frames	Percentage
idle	328	100.0%

9. Active AI Components

Component	Type	Status
Visual SLAM	Particle Filter + Occupancy Grid	ACTIVE (Phase 1)
YOLOv8-Nano	Deep Learning CNN	HEURISTIC (no ONNX)
Q-Learning	Reinforcement Learning	LOADED (epsilon=0.01)
A* Pathfinding	Classical AI Search	STANDBY (Phase 2)
Sensor Fusion	Bayesian Estimation	ACTIVE
Camera	Real Webcam (Laptop)	ACTIVE

10. Conclusion

The simulation ran successfully for 71.6 seconds with the laptop webcam providing live video frames. The AI obstacle detector processed 328 frames, detecting obstacles in 100.0% of frames. The robot traveled 1.18 meters during SLAM exploration. All safety systems functioned correctly with 0 tilt faults and 8 total bump contacts. Battery drain was minimal at 0.98% over the session.

The SLAM mapping phase did not complete (coverage < 85%) as the run duration was insufficient for full hospital coverage. A longer run (5-10 minutes) would allow the wall-following explorer to reach the coverage threshold and auto-save the map for Phase 2 navigation.